CLINICAL · MOUTH & THROAT

Nasopharyngeal carcinoma

Nasopharyngeal carcinoma hides in the one part of the throat no one sees without an endoscope. It shows itself instead through its neighbours: a lump high in the neck, a blocked ear with fluid in an adult, a bloodstained nose, a squint from a sixth nerve palsy. It is linked to Epstein–Barr virus, common in southern China and South-East Asia, and treated not with the knife but with radiation and chemotherapy, to which it responds well.

AT A GLANCE

- Adult with unilateral effusion or a neck node.
- Biopsy the fossa of Rosenmüller.
- Radiotherapy is the backbone.
- Surgery is for salvage.
- Induction gemcitabine–cisplatin for high risk; plasma EBV DNA for monitoring; cumulative cisplatin about 200 mg/m².

1 CHAPTER ONE Overview

A cancer of geography and virus.

1.1 Tumours of the nasopharynx

TUMOURS OF THE NASOPHARYNX

BENIGN	MALIGNANT
Juvenile angiofibroma (the commonest benign tumour)	Nasopharyngeal carcinoma (the commonest malignancy)
Antrochoanal polyp, squamous papilloma	Lymphoma (mostly B-cell non-Hodgkin)
Thornwaldt cyst	Rhabdomyosarcoma (children)
Pleomorphic adenoma, craniopharyngioma	Chordoma, plasmacytoma
Paraganglioma, hamartoma	Adenoid cystic and other salivary carcinomas

BENIGN	MALIGNANT
Hairy polyp, teratoma (congenital)	Melanoma, haemangiopericytoma

Angiofibroma has its own topic; this one is about carcinoma.

1.2 Epidemiology

The highest incidence in the world is in southern China, Hong Kong and Taiwan, with high rates in South-East Asia, North Africa, the Mediterranean and among the Inuit and Aleut peoples. It is uncommon in India except in the North-East. Chinese born in North America have a lower risk than those born in China but still higher than other Americans, showing the interplay of genes and environment. Males outnumber females about three to one; the peak is in the fifth and sixth decades, but about one in five patients is under thirty, and these younger patients usually have undifferentiated tumours.

1.3 Aetiology

- **Genetic:** HLA haplotypes confer susceptibility; the risk persists after migration.
- **Epstein–Barr virus:** EBV DNA and EBV-encoded RNA are found in virtually every tumour cell of the non-keratinising types; clonal EBV in premalignant lesions suggests infection is an early, possibly initiating, event.
- **Environmental:** nitrosamines in salted, preserved fish eaten from childhood, with a diet low in vitamin C; smoke from burning incense and wood; tobacco, opium and occupational fumes.
- **Tobacco and alcohol:** linked to the keratinising (sporadic) type.

2 CHAPTER TWO Pathology

Three types.

2.1 WHO classification

WHO TYPES OF NASOPHARYNGEAL CARCINOMA

TYPE	DESCRIPTION	NORTH AMERICAN FREQUENCY	EBV AND RADIOSENSITIVITY
I	Keratinising squamous cell carcinoma	About 25%	Weaker association; least radiosensitive
II	Non-keratinising, differentiated	About 12%	Strong association
III	Non-keratinising, undifferentiated (lymphoepithelioma; Regaud and Schmincke types)	About 63%	Strongest association; most radiosensitive

The current WHO scheme groups types II and III as non-keratinising carcinoma (differentiated and undifferentiated). Under the electron microscope all are squamous; immunohistochemistry for epithelial (cytokeratin) and lymphoid markers separates undifferentiated carcinoma from lymphoma. In endemic areas the non-keratinising types predominate even more.

2.2 Gross forms

Proliferative (a polypoid mass causing nasal obstruction), ulcerative (bleeding), and infiltrative (submucosal, easily missed).

3 CHAPTER THREE Spread

Through every exit of the skull base.

3.1 Local

The commonest site of origin is the *fossa of Rosenmüller* in the lateral wall.

- **Forwards:** into the choanae and nasal cavity.
- **Laterally:** through the sinus of Morgagni into the parapharyngeal space and infratemporal fossa; into the pterygoid muscles (trismus).
- **Upwards:** through the foramen lacerum and foramen ovale into the middle cranial fossa and cavernous sinus (cranial nerves III, IV, V and VI); the *sixth nerve* is usually the first affected; into the sphenoid sinus and orbit (proptosis, optic nerve).
- **Backwards:** to the jugular foramen (IX, X, XI) and hypoglossal canal (XII), and the cervical sympathetic chain (Horner syndrome), directly or through enlarged retropharyngeal nodes.
- **Downwards:** to the oropharynx.

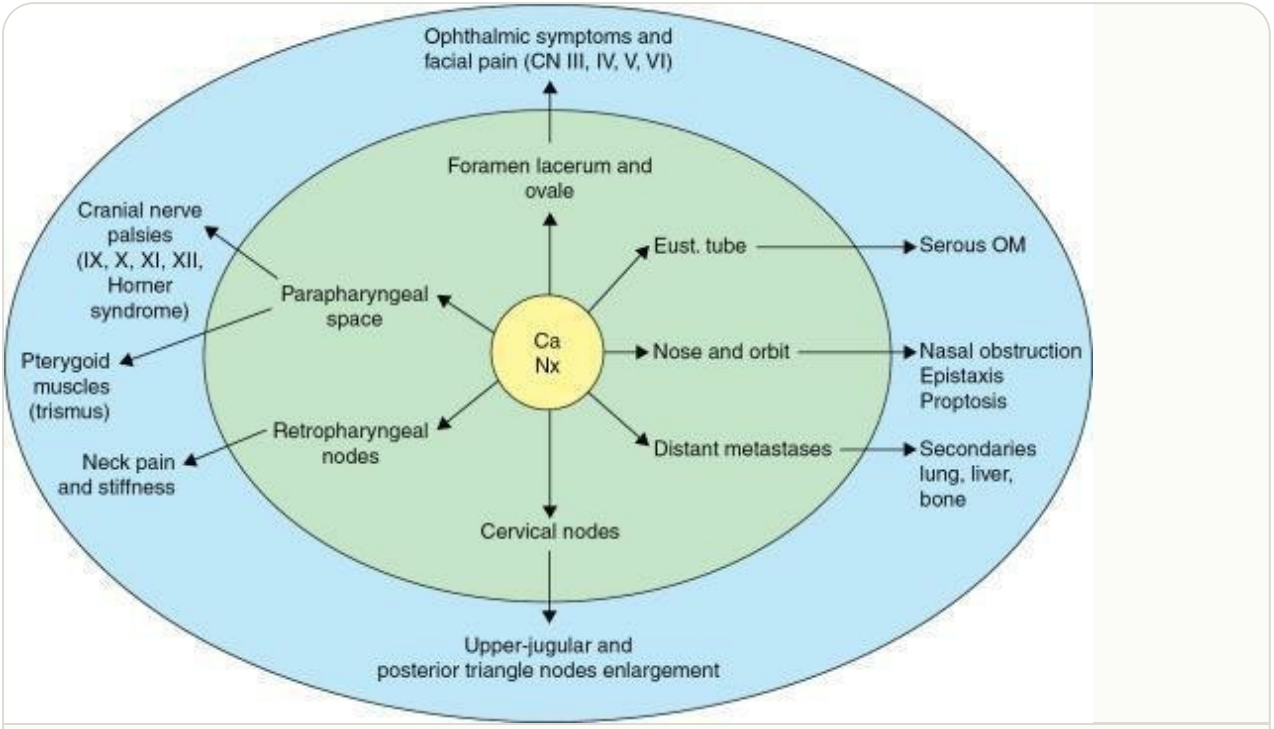


FIGURE 1 Routes of spread of nasopharyngeal carcinoma and the clinical features they produce. Dhingra, fig. 49.9

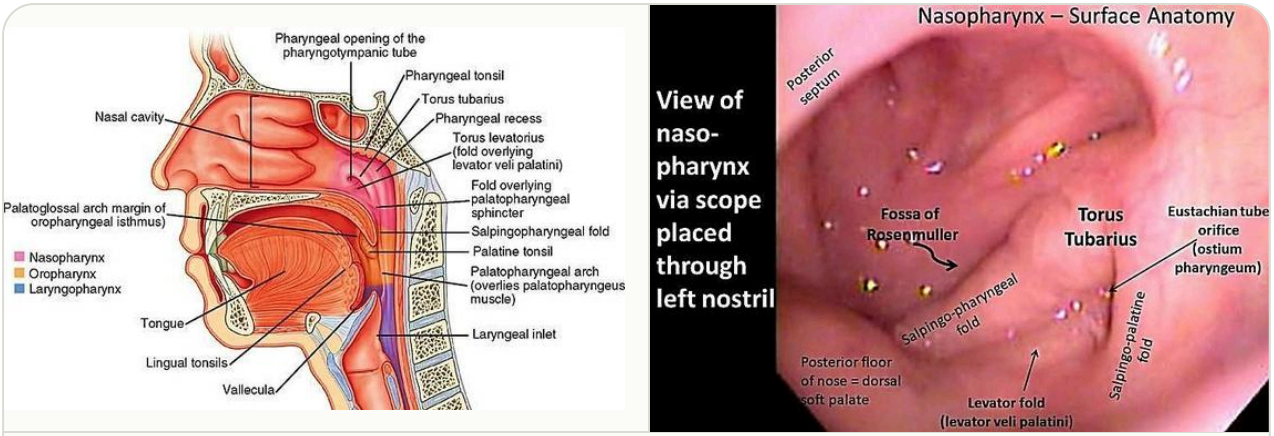


FIGURE 2 The lateral wall of the nasopharynx with the torus tubarius and fossa of Rosenmüller, and an endoscopic view.

Anatomy atlas · ENT Steps p. 203

3.2 Lymphatic

The nasopharynx is rich in lymphatics and nodal spread is early and often bilateral: to the lateral retropharyngeal node (of Rouvière), the upper jugular (level II) nodes high under the mastoid, and the spinal accessory chain in the posterior triangle (level V). Retropharyngeal nodes cause neck stiffness and press on the lower cranial nerves.

3.3 Distant

Bone, lung and liver. Nasopharyngeal carcinoma has the highest rate of distant metastasis of any head and neck cancer, present or developing in about a quarter to a third of patients.

4 CHAPTER FOUR Clinical features

The four groups of symptoms.

4.1 Cervical nodes

The commonest presentation (in 60 to 90%): a painless lump between the angle of the jaw and the mastoid, or in the posterior triangle. Nodes are present in about

three-quarters at diagnosis, half of them bilateral; often the node is the only complaint.

4.2 Ear

Eustachian tube obstruction causes conductive hearing loss, aural fullness and *otitis media with effusion*; tinnitus; rarely tumour grows up the tube.

UNILATERAL EFFUSION IN AN ADULT

An adult with fluid behind one drum and no obvious cause has a nasopharyngeal examination, and a biopsy if anything is seen. Inserting a grommet without looking at the nasopharynx misses cancers.

4.3 Nose

Nasal obstruction, bloodstained discharge or epistaxis, and hyponasal speech.

4.4 Eye and nerves

Diplopia and squint from a sixth nerve palsy; ophthalmoplegia (III, IV, VI); facial pain and numbness with loss of the corneal reflex (V); proptosis and visual loss; the jugular foramen syndrome (IX, X, XI); deviation of the tongue (XII); Horner syndrome; headache from skull base erosion.

TROTTER'S TRIAD

Conductive deafness from tubal blockage, ipsilateral temporoparietal neuralgia from the fifth nerve, and immobility of the soft palate on the same side from the vagus: the classic signs of a tumour invading the lateral nasopharynx and parapharyngeal space.

4.5 Others

Trismus, neck pain from retropharyngeal nodes, weight loss, and the effects of distant metastases.

PRESENTING FEATURES IN ORDER OF FREQUENCY

FEATURE	NOTES
Neck mass	Most common; often the only symptom
Hearing loss and aural fullness	Effusion from tubal obstruction
Nasal obstruction and epistaxis	Bloodstained post-nasal discharge
Cranial nerve palsy	Sixth nerve most often
Headache, earache, neck pain	Skull base and retropharyngeal involvement

CHAPTER FIVE

5 Diagnosis and staging

See it, sample it, map it.

5.1 Endoscopy and biopsy

Rigid or flexible nasendoscopy under local anaesthesia shows a proliferative, ulcerative or submucosal lesion, and allows biopsy. If nodes suggest the diagnosis but nothing is seen, take biopsies from the fossa of Rosenmüller and posterior wall under general anaesthesia, sometimes through a transpalatal exposure. Fine-needle aspiration of a neck node, with EBV testing of the cells, points to a nasopharyngeal primary.

5.2 Imaging

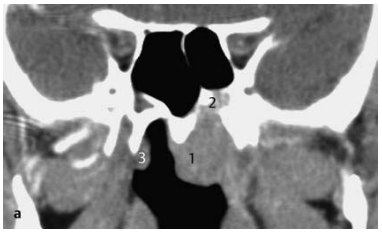
- **MRI:** the best study for soft tissue extent, the parapharyngeal and retropharyngeal spaces, perineural spread, skull base marrow, intracranial extension and nodes.
- **CT:** bone erosion of the skull base, clivus and pterygoid plates.
- **PET-CT:** distant metastases and detection of residual or recurrent disease after treatment; chest and abdominal imaging if PET is unavailable.

5.3 Other tests

A baseline *audiogram* (radiation and cisplatin both damage hearing), dental assessment before radiotherapy, and *EBV serology*: IgA antibodies to the early antigen are highly specific and those to the viral capsid antigen more sensitive, so both are used; plasma EBV DNA load is now used for screening in endemic areas, for prognosis and to monitor for recurrence.

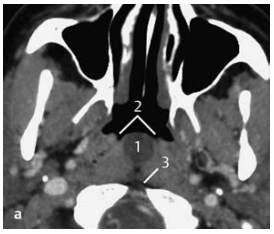
IMAGING IN THE APP 2 cases

Nasopharyngeal carcinoma and a Thornwaldt cyst on imaging.



Nasopharyngeal Carcinoma

Tumours · 11 images



Thornwaldt Cyst

Congenital · 2 images

5.4 Staging

TNM STAGING OF NASOPHARYNGEAL CARCINOMA (AJCC)

CATEGORY	DEFINITION
T1	Confined to the nasopharynx, or extending to the oropharynx or nasal cavity without parapharyngeal involvement
T2	Parapharyngeal extension, or involvement of the medial or lateral pterygoid or prevertebral muscles
T3	Invasion of bony structures of the skull base, cervical vertebrae or pterygoid structures, or the paranasal sinuses
T4	Intracranial extension, cranial nerves, hypopharynx, orbit, parotid, or soft tissue beyond the lateral pterygoid
N1	Unilateral cervical, or unilateral or bilateral retropharyngeal, nodes 6 cm or less, above the caudal border of the cricoid

CATEGORY	DEFINITION
N2	Bilateral cervical nodes 6 cm or less, above the caudal border of the cricoid
N3	Node over 6 cm, or extension below the caudal border of the cricoid

Dhingra quotes the older AJCC definitions, in which T1 was confined to the nasopharynx and N3b meant any node in the supraclavicular fossa (Ho's triangle, bounded by the medial and lateral ends of the clavicle and the point where neck meets shoulder). In every edition the nodal staging differs from other head and neck sites: low neck nodes carry the worst prognosis whatever their size.

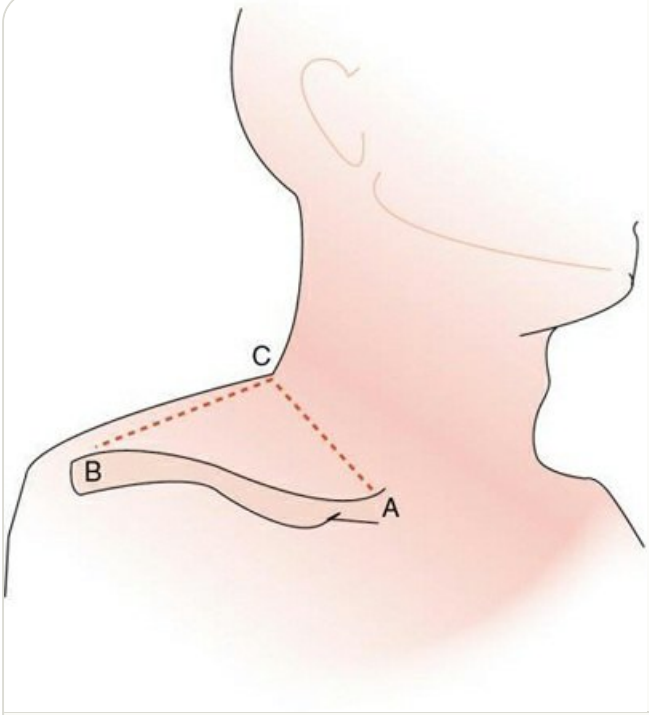


FIGURE 3 Ho's triangle (the supraclavicular fossa): the medial (A) and lateral (B) ends of the clavicle and the point where the neck meets the shoulder (C).
Dhingra, fig. 49.10

6 CHAPTER SIX Treatment

Radiation first.

6.1 Primary treatment

The nasopharynx is hard to reach surgically, and these tumours are radiosensitive, so *radiotherapy is the treatment of choice*.

- **Stage I:** radiotherapy alone to the nasopharynx and both sides of the neck, about 70 Gy.
- **Stage II to IVA:** concurrent chemoradiation with cisplatin, which improves survival over radiotherapy alone, with or without induction or adjuvant chemotherapy (cisplatin, fluorouracil, gemcitabine).
- **Intensity-modulated radiotherapy (IMRT)** delivers a high dose to the tumour while sparing the brainstem, spinal cord, optic apparatus, temporal lobes, cochleae and parotids.
- **Metastatic disease:** platinum-based chemotherapy, now with immunotherapy, and local treatment in selected cases.

6.2 Residual and recurrent disease

- **Neck:** persistent or recurrent nodes after radiation are treated by neck dissection; Dhingra advises a radical dissection because disease is often more extensive than it seems.
- **Nasopharynx:** after restaging with MRI and PET-CT, small recurrences are treated by *nasopharyngectomy*: endoscopic for limited central tumours, or open through a maxillary swing, transpalatal, Le Fort I, midfacial degloving or infratemporal fossa approach. Extension around the internal carotid, intracranially or widely into the parapharyngeal space contraindicates surgery.
- **Re-irradiation:** IMRT, proton therapy or brachytherapy (interstitial implants, gold grains), at greater risk of necrosis of the skull base and temporal lobe, cranial nerve palsies and carotid rupture.

OPERATIVE GUIDES IN THEATRE 3 operations

Salvage nasopharyngectomy approaches.

Endoscopic Nasopharyngectomy
Salvage nasopharyngectomy
Robotic-Assisted Nasopharyngectomy
Robotic salvage
Maxillary Swing/Transpalatal/Midface Degloving With Le Fort I Osteotomy Approach for Nasopharyngectomy
Maxillary swing

6.3 Effects of treatment

Xerostomia and dental caries, otitis media with effusion and sensorineural hearing loss, trismus, hypothyroidism and pituitary failure, osteoradionecrosis of the skull base, temporal lobe necrosis, cranial neuropathy, carotid stenosis, and second cancers. Grommets in irradiated ears often discharge chronically.

6.4 Outcome

With concurrent chemoradiation, local and regional control is about 70 to 80% and five-year survival about 70 to 80%; distant metastasis is now the commonest mode of failure.

7 CHAPTER SEVEN Other malignancies

Not all are carcinoma.

- **Lymphoma:** a bulky submucosal mass, usually B-cell non-Hodgkin, often with disease elsewhere in Waldeyer's ring; sinonasal NK/T-cell lymphoma in Asian patients.
- **Rhabdomyosarcoma:** the embryonal type in children, as a polypoid mass with rapid growth and cranial nerve involvement; chemotherapy and radiotherapy.
- **Chordoma:** from notochordal remnants in the clivus, destroying bone and presenting in the nasopharynx; surgery and proton therapy.
- **Plasmacytoma, adenoid cystic carcinoma and melanoma** are rare.

CHEMOTHERAPY AND FOLLOW-UP (OXFORD TEXTBOOK)

Nonkeratinising, EBV-associated carcinoma is the commonest type everywhere; EBV is shown in tissue by EBER in-situ hybridisation. Distant metastases are present in about 4% at diagnosis and develop in about a quarter after IMRT, which is now the main cause of

treatment failure. For locoregionally advanced disease, concurrent cisplatin-based chemoradiotherapy improves survival over radiotherapy alone; the cumulative cisplatin dose matters more than the schedule, with about 200 mg/m² as the effective threshold (160 mg/m² after induction). **Induction gemcitabine and cisplatin** before chemoradiotherapy improves survival in high-risk patients, and **adjuvant or metronomic capecitabine** improves failure-free survival in high-risk disease; the benefit of conventional adjuvant chemotherapy remains debated. **Plasma EBV DNA** is a promising marker for screening high-risk populations, risk stratification and surveillance, but assays still need standardisation. Proton and carbon-ion therapy have dosimetric advantages without proven better outcomes yet. Follow-up is every three to four months for two years, six-monthly to five years, then yearly, with yearly CT or MRI for five years. Isolated nodal recurrence is treated by salvage neck dissection, and small local recurrences by endoscopic nasopharyngectomy or re-irradiation.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Aetiology:** EBV, genetics and salted fish.
- **WHO types** and their link to EBV and radiosensitivity.
- **Routes of spread** and the cranial nerves involved; **Trotter's triad.**
- **Unilateral effusion in an adult.**
- **Investigations:** endoscopy and biopsy, MRI, EBV serology and DNA.
- **Treatment:** radiotherapy, concurrent chemoradiation, salvage surgery.

8.2 Common pitfalls

COMMON PITFALLS

- Inserting a grommet in an adult with a one-sided effusion without examining the nasopharynx.
- Excising a high level II neck node before endoscopy and needle aspiration.

- Missing a sixth nerve palsy as the sign of skull base invasion.

Written for this app from Oxford Textbook of Otolaryngology ch. 61, Oxford Specialist Handbook ch. 20, Dhingra ch. 49 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Nasopharyngeal Carcinoma

IMAGING FINDINGS

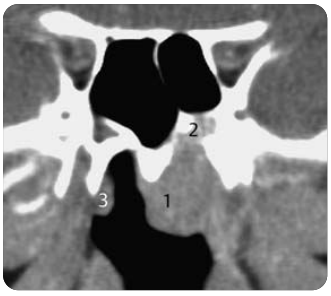
- Asymmetric mass in the fossa of Rosenmüller
- Parapharyngeal and skull base spread; clival marrow replaced on T1
- Retropharyngeal and level II and V nodes

TOP DIFFERENTIALS

- Adenoidal lymphoid hyperplasia (symmetric, striated)
- Lymphoma
- Minor salivary gland tumour

CHECKLIST

- MRI for staging: skull base, cavernous sinus, perineural spread
- Middle ear effusion from Eustachian tube obstruction



CT, coronal with intravenous contrast enhancement. Note the soft-tissue mass in the roof of the nasopharynx of the left side (1), with invasion in the…

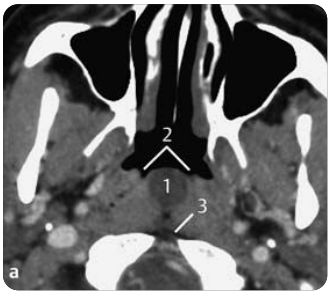


Coronal NECT of paranasal sinuses, performed in a 62-year-old Asian man for evaluation of nasal stuffiness and epistaxis, demonstrates an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484.

PHARYNX, TONSILS & OESOPHAGUS · CONGENITAL

Thornwaldt Cyst



CT, axial. During endoscopy, a rounded expansion on the posterior wall of the nasopharynx in a patient with sinonasal symptoms was noted. CT…